Type-A Poset Cones and Extension Tilers

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Abstract

We study chamber-unions in the Coxeter complex of type A_{n-1} through the braid-arrangement dictionary between reflecting halfspaces and precedence constraints. Nonempty halfspace-convex chamber-unions are exactly poset cones: the chambers satisfying a partial order P are the linear extensions $L(P)$. This turns Coxeter-pure group-translate tiling questions into exact factorizations of the symmetric group by sets of linear extensions. We recall and isolate the general positive theorem that every finite series-parallel poset is an extension tiler. We then give a finite classification proof for posets on five elements: among the 63 unlabeled types, the extension tilers in S_5 are exactly the 48 series-parallel types. The three non-series-parallel types for which $|L(P)|$ nevertheless divides 5! are excluded by elementary fiber-counting obstructions. A static supplementary audit file records the finite S_5 table and the former rank/packing certificates; the proof itself uses only the finite table and elementary obstructions. We formulate and verify an explicit six-element counterexample to the converse: a non-series-parallel poset whose linear extensions tile S_6. The counterexample has a compact double-coset multiplier set $A = HK \sqcup H\tau K$ with $H \cong D_4$ and $K \cong S_3$. Through the standard triangulation of the unit cube, this same factorization gives a coordinate-permutation tiling of $[0,1]^6$ by 48 copies of the order polytope of the counterexample, up to shared faces of the standard triangulation. Finally, we record an audit-level six-element laboratory table. It is not promoted to a theorem here: five negative audit rows remain supported only by bulky finite audit data, pending short independently checkable certificates.

1 Introduction

In a companion manuscript on the regular tetrahedron [Bab26], the A_3 Coxeter dissection is controlled by a small but rigid combinatorial object: posets on four elements. In that setting, convex chamber-unions are poset cones, the chamber count is the number of linear extensions, and the admissible convex tiling pieces are exactly the series-parallel poset types. The purpose of this note is to extract the type-A mechanism from that rank-three situation and ask what remains true in S_n. We use standard terminology for finite posets and linear extensions as in Trotter [Tro92] and Stanley [Sta12].

The general dictionary is simple. In the braid arrangement of type A_{n-1}, choosing a side of the reflecting hyperplane $x_i = x_j$ is the same as imposing one precedence constraint, either $i < j$ or $j < i$. Compatible systems of such choices close transitively to partial orders. The chambers satisfying those constraints are exactly the linear extensions of the resulting poset. Thus the halfspace-convex chamber unions in type A are naturally the poset cones $C(P)$.

The tiling question is more delicate. If P is a poset on $[n]$, write $L(P) \subseteq S_n$ for its linear extensions. We say that P is an extension tiler when

$$S_n = \bigsqcup_{a \in A} aL(P)$$

for some set $A \subseteq S_n$. This is a group-translate condition inside the Coxeter complex. It should not be confused with a classification of all possible Euclidean congruent copies in an ambient geometric problem. The setting is adjacent to earlier work on quotients and splittings in Coxeter groups [BW88, GG20], but differs from it because the complement set A is allowed to be an arbitrary set of left translators rather than a canonical weak-order quotient.

The first main point is positive and general: every finite series-parallel poset is an extension tiler. This is the mechanism already used in the A_3 Coxeter-pure paper; here we state it as a standalone type-A theorem.

The second main point begins at $n = 5$. In rank four, divisibility stops being a sufficient proxy for series-parallelity. There are 51 unlabeled poset types on five elements for which $|L(P)|$ divides 120, but only 48 of them are series-parallel. We prove that the remaining three types do not tile S_5 by left translates using elementary fiber-counting obstructions. The former rank and packing certificates are retained only as an independent computational audit.

The resulting general theorem is only one-sided:

$$\text{series-parallel} \Longrightarrow \text{extension tiler}$$

for all finite posets. The converse is known in the previously analyzed A_3/S_4 Coxeter-pure setting and is proved here for S_5:

$$\text{series-parallel} \Longleftrightarrow \text{extension tiler}$$

in those finite test cases. However, the next rank already produces a counterexample: we give a six-element non-series-parallel poset whose linear extensions tile S_6.

The same exact factorization also has a geometric interpretation. Under the standard simplex triangulation of the unit cube, the order polytope $\mathcal{O}(P)$ is the union of the simplices indexed by $L(P)$. Thus an extension-tiler factorization of S_n produces a tiling of $[0,1]^n$ by coordinate-permuted copies of $\mathcal{O}(P)$. For the six-element counterexample this gives a tiling of $[0,1]^6$ by 48 copies of its order polytope.

We keep three claim levels separate throughout the paper. The type-A poset-cone dictionary and the series-parallel tiler theorem are ordinary theorems. The S_5 classification is also an ordinary finite theorem, with the computer relegated to audit status. The S_6 example is a finite certificate that the naive converse is false; the remaining problem is to classify the larger class of extension tilers.

2 Poset Cones in Type A

Let X be a finite label set with $|X| = n$. We write S_X for the set of all bijective words on X, identified with the symmetric group after choosing an ordering of X.

The braid arrangement of type A_{n-1} has reflecting hyperplanes

$$x_i = x_j \qquad (i \neq j).$$

Its chambers are total orders of the coordinates, hence are indexed by permutations; see, for example, the standard treatment of Coxeter groups in [BB05]. We use the convention that a word

$$\sigma = (\sigma_1, \dots, \sigma_n)$$

records the labels from first to last in the corresponding total order.

Definition 2.1. Let P be a strict partial order on X. The set of linear extensions of P is

$$L(P) = \{\sigma \in S_X : \mathrm{pos}_\sigma(a) < \mathrm{pos}_\sigma(b) \text{ whenever } a <_P b\}.$$

The associated poset cone is the chamber-union

$$C(P) = L(P).$$

Here the equality $C(P) = L(P)$ means that we identify chamber-unions with their chamber-indexing sets. Geometrically, the relation $a <_P b$ selects one side of the reflecting hyperplane $x_a = x_b$.

The correspondence between poset cones and unions of chambers of the braid arrangement is classical: poset cones and their Whitney-number refinements of the linear-extension count are developed in [DBKR21], and a comprehensive treatment of the faces and cones of a real hyperplane arrangement is given in [AM17]. These are context references; no result from them is used in the proofs below.

Theorem 2.2 (Type-A halfspace poset-cone dictionary)**.** *In the Coxeter complex of type A_{n-1}, the nonempty chamber-unions defined by intersections of reflecting halfspaces are exactly the poset cones $C(P)$ for labeled posets P on X.*

Proof. A reflecting halfspace choice for $x_i = x_j$ is exactly one precedence constraint, either $i < j$ or $j < i$. If a system of such choices is nonempty, then the corresponding precedence constraints are acyclic; their transitive closure is a partial order P. The chambers satisfying the original halfspace choices are exactly the total orders extending this partial order, namely $L(P)$.

Conversely, every poset P gives an intersection of reflecting halfspaces by imposing $a < b$ for each relation $a <_P b$. The chambers in that intersection are precisely the linear extensions of P. ☐

Proposition 2.3 (Orbit dictionary)**.** *The natural action of S_X on type-A chambers sends poset cones to poset cones by relabeling. Hence the S_X-orbits of poset cones are in bijection with isomorphism classes of posets on X.*

Proof. Applying a permutation $\tau \in S_X$ to a chamber word relabels every entry. Thus the linear extensions of P are sent to the linear extensions of the relabeled poset τP. This proves equivariance. Two labeled posets lie in the same relabeling orbit exactly when they are isomorphic. ☐

Proposition 2.4 (Connectivity)**.** *Every poset cone $C(P)$ is connected in the chamber graph.*

Proof. The chamber graph of type A is the adjacent-transposition graph on words. The graph of linear extensions of a finite poset is connected [Sta12, Sec. 3.5]; see also the generation algorithm of Pruesse and Ruskey [PR94]: given two extensions, one can move from one to the other by swapping adjacent incomparable elements. These swaps are precisely chamber-graph adjacencies that remain inside $C(P)$. ☐

Remark 2.5. The theorem in this section is deliberately stated for halfspace convexity. In the A_3 tetrahedral setting, geometric, geodesic, and halfspace convexity coincide for connected chamber-unions. For arbitrary type A_{n-1}, that stronger equivalence is not used in this paper unless supplied by a separate proof or reference.

3 Extension Tilers

Definition 3.1. Let P be a poset on a finite label set X. We say that P is an *extension tiler* if there exists a set $A \subseteq S_X$ such that

$$S_X = \bigsqcup_{a \in A} aL(P).$$

This definition packages the Coxeter-pure group-translate tiling problem as an exact factorization problem in the symmetric group. The translates are left translates because the Coxeter group acts on chambers by left composition. Related tiling questions in symmetric groups can be phrased through partition-transitivity and permutation modules; see Martin–Sagan for the general notion of λ-transitivity [MS06] and Fang–Xia for a recent symmetric-group tiling problem where such conditions are used explicitly [FX26].

Lemma 3.2 (Relabeling invariance)**.** *The extension-tiler property is invariant under relabeling of the ground set.*

Proof. Let $\tau : X \to Y$ be a bijection. Relabeling sends $L(P)$ to $L(\tau P)$ and conjugates the corresponding left-translate family from S_X to S_Y. Therefore a disjoint cover of S_X by translates of $L(P)$ gives a disjoint cover of S_Y by translates of $L(\tau P)$. □

There are two equivalent finite formulations that will be used for certificates.

Definition 3.3 (Exact-cover matrix)**.** Fix P on $[n]$. Let the columns be the distinct left translates $gL(P)$ and let the rows be the elements $\sigma \in S_n$. Define

$$M_{\sigma,gL(P)} = \begin{cases} 1, & \sigma \in gL(P), \\ 0, & \text{otherwise.} \end{cases}$$

Then P is an extension tiler if and only if the integer system

$$Mx = \mathbf{1}, \qquad x \in \{0,1\}^{\#\text{columns}}$$

has a solution.

Reducing this system modulo a prime can prove nonexistence. In Appendix B, the two former $|L(P)| = 5$ audit certificates are recorded as $\mathbb{F}_5$-rank inconsistencies; the main S_5 proof below uses elementary fiber-counting obstructions instead. The underlying enumeration is intentionally finite and small; in general, counting linear extensions is computationally difficult [BW91].

Definition 3.4 (Packing graph)**.** The packing graph of P has as vertices the distinct left translates of $L(P)$. Two vertices are adjacent if the corresponding translates are disjoint.

If $|L(P)|$ divides $n!$, a tiling requires a clique of size $n!/|L(P)|$ in this graph. Since left translation acts transitively on the translate family, any tiling can be translated so that it contains the base tile $L(P)$. Thus it is enough to search for a clique of the required size containing the base translate.

Remark 3.5. The word "tiler" in this paper always means extension tiler, i.e. a left-translate exact factorization of a symmetric group by a set of linear extensions. It is not an assertion about arbitrary ambient Euclidean congruences.

4 Series-Parallel Posets Tile

Let P and Q be posets on disjoint ground sets. Their *series composition* $P;Q$ keeps the internal relations of P and Q and adds $p < q$ for every $p \in P$ and $q \in Q$. Their *parallel composition* $P\|Q$ keeps the internal relations and adds no cross-relations. A finite poset is series-parallel if it is obtained from one-element posets by iterating these two operations. Equivalently, finite series-parallel posets are the N-free posets: this is the Valdes–Tarjan–Lawler characterization applied to the transitive directed comparability graph of the poset [VTL82]. The linear-extension structure of N-free orders is studied further in Felsner–Manneville [FM15].

Lemma 4.1 (Extension counts)**.** *If $|P| = p$ and $|Q| = q$, then*

$$|L(P;Q)| = |L(P)|\,|L(Q)|$$

and

$$|L(P\|Q)| = \binom{p+q}{p}|L(P)|\,|L(Q)|.$$

Proof. For $P; Q$, every element of P must precede every element of Q, so a linear extension is an extension of P followed by an extension of Q.

For $P \| Q$, choose the p positions occupied by the elements of P, fill them by a linear extension of P, and fill the remaining positions by a linear extension of Q. □

Lemma 4.2 (Series closure)**.** *If P_1 and P_2 are extension tilers, then $P_1; P_2$ is an extension tiler.*

Proof. Let $|P_1| = p$, $|P_2| = q$, and let X be a label set of size $p + q$. Given a word $\sigma \in S_X$, let Y be the set of the first p letters of σ and let $Z = X \setminus Y$.

In a linear extension of a series composition, the label set of the first component is forced to appear before the label set of the second component. Therefore Y is forced for any tile containing σ. Restrict σ to its subwords on Y and on Z. By relabeling invariance and the extension-tiler property of P_1 and P_2, these subwords lie in unique component tiles. Combining those two component tiles gives a tile for a relabeling of $P_1; P_2$ containing σ.

The forced block Y and uniqueness inside the two component tilings give disjointness. Hence the constructed family is a partition of S_X. □

Lemma 4.3 (Parallel closure)**.** *If P_1 and P_2 are extension tilers, then $P_1 \| P_2$ is an extension tiler.*

Proof. Let $|P_1| = p$, $|P_2| = q$, and fix one decomposition $X = Y \sqcup Z$ with $|Y| = p$ and $|Z| = q$. Given $\sigma \in S_X$, read the subword $\sigma|_Y$ and the subword $\sigma|_Z$. By relabeling the component tilings to the arbitrary label sets Y and Z, these determine unique relabeled tiles for P_1 on Y and for P_2 on Z.

Since parallel composition adds no cross-relations, every shuffle of a linear extension from the first component with a linear extension from the second component is a linear extension of the parallel composition. Thus the two component tiles determine a tile of $P_1 \| P_2$ containing σ. Disjointness follows from the uniqueness of the two subwords' component tiles. □

Theorem 4.4 (Series-parallel extension-tiler theorem)**.** *Every finite series-parallel poset is an extension tiler.*

Proof. The one-element poset is an extension tiler. By Lemmas 4.2 and 4.3, the class of extension tilers is closed under series and parallel composition. Therefore every poset built from singletons by these operations is an extension tiler. □

Corollary 4.5. *If P is series-parallel on n elements, then $|L(P)|$ divides $n!$.*

Proof. By Theorem 4.4, S_n is a disjoint union of translates of $L(P)$, so $|L(P)|$ divides $|S_n| = n!$. □

Remark 4.6. The converse divisibility statement is false for $n = 5$: there are non-series-parallel posets on five elements with $|L(P)|$ dividing 120. The next section certifies that these divisible false positives still do not tile by left translates.

5 The S_5 Classification

The first rank where divisibility stops characterizing the series-parallel types is $n = 5$. We treat S_5 as the first nontrivial test of the extension-tiler converse.

We first isolate the finite classification input. Let N be the four-element poset on $\{a, b, c, d\}$ with relations

$$a < b, \qquad c < b, \qquad c < d.$$

This is the forbidden induced poset in the Valdes–Tarjan–Lawler characterization of finite series-parallel posets, viewing a poset through its transitive directed comparability graph.

Lemma 5.1 (One-point N-extensions)**.** *Every non-series-parallel poset on five elements is isomorphic to a one-point extension of N.*

More explicitly, let x be the added point and put

$$D = \{y \in N : y < x\}, \qquad U = \{y \in N : x < y\}.$$

Then D is an order ideal of N, U is an order filter of N, and

$$d <_N u \quad \textit{for every } d \in D,\ u \in U.$$

Conversely, any pair (D, U) satisfying these three conditions defines a five-element poset containing N as an induced subposet.

Proof. By the Valdes–Tarjan–Lawler characterization, a finite poset is non-series-parallel exactly when it contains an induced N-poset. Thus a non-series-parallel poset on five elements is obtained from an induced copy of N by adding one further point x.

If $y < x$ and $z < y$ in N, then $z < x$, so D is an ideal. Dually, if $x < y$ and $y < z$ in N, then $x < z$, so U is a filter. Finally, if $d \in D$ and $u \in U$, then $d < x < u$, hence $d < u$ by transitivity. Since the chosen four-point subposet is induced, this comparability must already belong to N.

Conversely, if D is an ideal, U is a filter, and every $d \in D$ is below every $u \in U$ in N, then adjoining relations $d < x$ for $d \in D$ and $x < u$ for $u \in U$ creates no new comparability among the four old elements beyond those already in N. Hence N remains an induced subposet. $\square$

Lemma 5.2 (Profile table)**.** *Up to isomorphism, there are exactly* 15 *non-series-parallel posets on five elements. Their numbers of linear extensions are*

$$5, 5, 7, 7, 7, 8, 9, 9, 11, 14, 14, 16, 16, 18, 18, 25.$$

Proof. The ideals of N are

$$\emptyset,\ a,\ c,\ ac,\ cd,\ abc,\ acd,\ abcd,$$

and the filters of N are

$$\emptyset,\ b,\ d,\ ab,\ bd,\ abd,\ bcd,\ abcd.$$

Here, for instance, ac denotes the set $\{a, c\}$.

The valid profiles (D, U) from Lemma 5.1 are exactly the nonempty entries of the following matrix. Each entry is the value of $|L(P_{D,U})|$ for the corresponding one-point extension:

$D\backslash U$	$\emptyset$	b	d	ab	bd	abd	bcd	$abcd$
$\emptyset$	25	18	16	9	14	8	7	5
a	16	9						
c	18	11	9		7			
ac	14	7						
cd	9							
abc	7							
acd	8							
$abcd$	5							

The entries are computed by insertion into the five linear extensions of N:

$$acbd, \quad acdb, \quad cabd, \quad cadb, \quad cdab.$$

For a profile (D, U), the number $|L(P_{D,U})|$ is the sum, over these five words, of the number of slots in which x may be inserted after all elements of D and before all elements of U.

There are 20 valid profiles before quotienting by isomorphism. The only identifications among them are

$$(ac, b) \cong (abc, \emptyset), \qquad (\emptyset, bcd) \cong (c, bd),$$
$$(\emptyset, abd) \cong (acd, \emptyset), \qquad (c, d) \cong (cd, \emptyset),$$
$$(\emptyset, ab) \cong (a, b).$$

One may check these five equivalences by the following explicit relabelings of $\{a, b, c, d, x\}$, written in cycle notation:

profiles	relabeling
$(ac, b) \cong (abc, \emptyset)$	$(b\,x)$
$(\emptyset, bcd) \cong (c, bd)$	$(c\,x)$
$(\emptyset, abd) \cong (acd, \emptyset)$	$(a\,d\,b\,x\,c)$
$(c, d) \cong (cd, \emptyset)$	$(d\,x)$
$(\emptyset, ab) \cong (a, b)$	$(a\,x)$.

Appendix A records a short invariant check that there are no further identifications: for each of the 15 quotient classes, the pair consisting of $|L|$ and the multiset of lower/upper comparability counts of its elements is distinct.

Thus the 20 profiles give exactly 15 isomorphism classes. Reading one representative from each class gives the displayed multiset of extension numbers. □

Corollary 5.3 (Divisibility reduction)**.** *Among non-series-parallel posets on five elements, only three isomorphism classes satisfy $|L(P)| \mid 120$: the two classes with $|L(P)| = 5$ and the single class with $|L(P)| = 8$.*

Lemma 5.4 (Fixed end-position obstruction)**.** *Let P be a poset on five elements with $|L(P)| = 5$. If all linear extensions of P have the same first letter, or all have the same last letter, then P is not an extension tiler.*

Proof. Suppose first that every word in $L(P)$ begins with p. For any $a \in S_5$, every word in $aL(P)$ begins with $a(p)$, so the translate is contained in one first-letter fiber

$$F_x^{\text{first}} = \{\sigma \in S_5 : \sigma_1 = x\}.$$

Each such fiber has size $4! = 24$. A tiling by translates of $L(P)$ would partition every first-letter fiber into blocks of size 5, impossible because $5 \nmid 24$.

The last-letter case is identical, using the fibers $F_x^{\text{last}} = \{\sigma \in S_5 : \sigma_5 = x\}$. □

Lemma 5.5 (Two-letter obstruction)**.** *Let $P = P_{\emptyset,abd}$ be the one-point extension with relations*

$$a < b, \qquad c < b, \qquad c < d, \qquad x < a, \qquad x < b, \qquad x < d.$$

Then P is not an extension tiler.

Proof. The linear extensions of P are

$$cxabd, \quad cxadb, \quad cxdab, \quad xacbd, \quad xacdb,$$
$$xcabd, \quad xcadb, \quad xcdab.$$

Thus their first-letter and first-two-letter distributions are

$$\#\{\sigma \in L(P) : \sigma_1 = x\} = 5, \qquad \#\{\sigma \in L(P) : \sigma_1 = c\} = 3,$$

and

$$\#(xc) = 3, \qquad \#(cx) = 3, \qquad \#(xa) = 2,$$

where $\#(uv)$ denotes the number of extensions beginning with the ordered pair (u, v).

Assume for contradiction that S_5 is partitioned by 15 left translates of $L(P)$. For each translate $gL(P)$ write

$$g(x) = u, \qquad g(c) = v, \qquad g(a) = w.$$

Then this translate contributes 3 elements to the ordered-pair fiber (u, v), 3 elements to the fiber (v, u), and 2 elements to the fiber (u, w).

First count only first letters. Let

$$\alpha_y = \#\{g : g(x) = y\}, \qquad \beta_y = \#\{g : g(c) = y\}.$$

The first-letter fiber over y has size 24, so

$$5\alpha_y + 3\beta_y = 24.$$

The nonnegative integer solutions are $(\alpha_y, \beta_y) = (3, 3)$ and $(0, 8)$. Since $\sum_y \alpha_y = 15$, the second possibility cannot occur. Thus

$$\alpha_y = \beta_y = 3$$

for every label y.

Now fix distinct labels u, v, and define

$$N_{uv} = \#\{g : g(x) = u,\ g(c) = v\}, \qquad C_{uv} = \#\{g : g(x) = u,\ g(a) = v\}.$$

The ordered-pair fiber (u, v) has size $3! = 6$, hence

$$3N_{uv} + 3N_{vu} + 2C_{uv} = 6.$$

Modulo 3, this gives $C_{uv} \equiv 0 \pmod 3$. For fixed u,

$$\sum_{v \neq u} C_{uv} = \alpha_u = 3,$$

so there is a unique value $f(u) \neq u$ with $C_{u,f(u)} = 3$, and all other C_{uv} are zero.

The fiber equation now says that if $f(u) = v$, then $N_{uv} + N_{vu} = 0$, while if $f(u) \neq v$, then $N_{uv} + N_{vu} = 2$. Applying the same statement to the opposite ordered pair (v, u) gives

$$f(u) = v \quad \Longleftrightarrow \quad f(v) = u.$$

Thus f would be a fixed-point-free involution on a set of five elements. No such involution exists. This contradiction excludes the tiling. $\square$

Theorem 5.6 (S_5 classification)**.** *For posets P on five elements, $L(P)$ tiles S_5 by left translates if and only if P is series-parallel.*

Proof. The positive direction follows from Theorem 4.4. Conversely, suppose that P is an extension tiler on five elements. Then $|L(P)|$ must divide $|S_5| = 120$.

If P is not series-parallel, Corollary 5.3 shows that only three non-series-parallel types pass this divisibility test: the two types with $|L(P)| = 5$ and the type with $|L(P)| = 8$.

The two $|L(P)| = 5$ profiles are $(abcd, \emptyset)$ and $(\emptyset, abcd)$. In the first, all linear extensions end with x; in the second, all linear extensions begin with x. Lemma 5.4 excludes both types.

The remaining divisible non-series-parallel type is represented by $P_{\emptyset,abd}$ and is excluded by Lemma 5.5. Therefore no non-series-parallel poset on five elements is an extension tiler. $\square$

For auditability, the finite table above and the former rank and packing certificates are mirrored in the static audit file

`certificates/s5_expected_output.txt`.

The theorem itself uses only the finite table and the elementary obstructions proved above. The supplementary replay script is included for auditability and reproduces the aggregate table together with the former rank and packing certificates; it is not a dependency of the proof.

Remark 5.7. This $n = 5$ means Coxeter type A_4 and the group S_5. It is not the separate geometric question of dissecting a tetrahedron into five congruent pieces.

6 A Counterexample in S_6

The S_5 classification suggests the natural converse to Theorem 4.4: perhaps every extension tiler is series-parallel. This converse is false in the next rank.

Theorem 6.1 (A non-series-parallel extension tiler in S_6)**.** *Let P be the six-element poset with cover relations*

$$0 < 1, \qquad 1 < 2, \qquad 1 < 3, \qquad 4 < 3, \qquad 4 < 5, \qquad 5 < 2.$$

Then P is not series-parallel, but $L(P)$ tiles S_6 by left translates.

Proof. The poset contains induced copies of the forbidden N-poset; for example, the induced subposet on $\{1, 2, 3, 5\}$ has relations

$$1 < 2, \qquad 1 < 3, \qquad 5 < 2,$$

which is isomorphic to N under

$$a \mapsto 5, \qquad b \mapsto 2, \qquad c \mapsto 1, \qquad d \mapsto 3.$$

Hence P is not series-parallel.

We now give a compact multiplier set. Let permutations act on the labels of words, so that

$$g(w_1, \ldots, w_6) = (g(w_1), \ldots, g(w_6)).$$

Define

$$H = \langle (2\,3)(4\,5),\ (0\,1)(3\,5) \rangle,$$
$$K = \langle (2\,3)(4\,5),\ (0\,4\,5)(1\,3\,2) \rangle,$$

and let $\tau = (3\,4)$. Put

$$A = HK \sqcup H\tau K.$$

A direct computation gives

$$|L(P)| = 15, \qquad |H| = 8, \qquad |K| = 6,$$
$$|HK| = |H\tau K| = 24, \qquad HK \cap H\tau K = \emptyset.$$

Thus $|A| = 48$, which is the required number of tiles since $48 \cdot 15 = 720 = |S_6|$.

It remains only to check disjointness. The verification is finite: compute the 15 linear extensions of P, form all words $a\sigma$ with $a \in A$ and $\sigma \in L(P)$, and count multiplicities. The resulting 720 words are all distinct. Hence

$$S_6 = \bigsqcup_{a \in A} aL(P),$$

so P is an extension tiler. $\square$

The verification in the last paragraph is intentionally small: it uses only the two generated subgroups H, K and the element τ, rather than a list of 48 multipliers. The supplementary script

`scripts/verify_s6_p038_biset_counterexample.py`

reproduces the calculation and also confirms that this poset contains four induced copies of N.

Corollary 6.2. *The converse of Theorem 4.4 is false.*

Corollary 6.3 (Coxeter-complex realization). *Let P be the six-element poset of Theorem 6.1. In the type A_5 Coxeter complex, whose chambers are indexed by the elements of S_6, the chamber set $L(P)$ is a 15-chamber Coxeter-pure piece. The 48 left translates*

$$\{aL(P) : a \in A\}$$

from Theorem 6.1 form a disjoint partition of the 720 Coxeter chambers. Equivalently, the verified P038 factorization realizes a type A_5 Coxeter-pure group-translate tiling by 48 translates of the 15-chamber poset cone $C(P) = L(P)$.

Proof. Index the chambers of the type A_5 Coxeter complex by words $w \in S_6$. The chamber graph is the adjacent-transposition graph on words. Relabeling every entry of a word by an element of S_6 sends adjacent words to adjacent words, and therefore preserves the Coxeter chamber structure.

Theorem 6.1 gives the exact factorization

$$S_6 = \bigsqcup_{a \in A} aL(P),$$

where $|L(P)| = 15$ and $|A| = 48$. Replacing each word by the corresponding Coxeter chamber gives a disjoint cover of all 720 type A_5 chambers by 48 left translates of the 15-chamber set $L(P)$. Since each translate is obtained by a Coxeter-group action, the pieces are congruent as Coxeter-chamber unions. □

Remark 6.4. The multiplier set above has visible two-sided structure: it is a union of two (H, K)-double cosets, with $H \cong D_4$ and $K \cong S_3$. This suggests that non-series-parallel extension tilers in S_6 are not merely sporadic exact-cover artefacts; at least some arise from structured group-theoretic complements.

7 Order-Polytope Cube Tilings

We record the geometric shadow of an extension-tiler factorization. This section uses only the standard order polytope of a poset and the standard simplex triangulation of the cube; no projected or visual model is part of the proof.

Let P be a poset on a finite label set X, with $|X| = n$. Its order polytope is

$$\mathcal{O}(P) = \{x \in [0,1]^X : x_i \leq x_j \text{ whenever } i <_P j\},$$

in the sense of Stanley [Sta86]. For a word $\sigma = (\sigma_1, \ldots, \sigma_n) \in S_X$, let

$$\Delta_\sigma = \{x \in [0,1]^X : 0 \leq x_{\sigma_1} \leq x_{\sigma_2} \leq \cdots \leq x_{\sigma_n} \leq 1\}.$$

The closed simplices Δ_σ, for $\sigma \in S_X$, form the standard triangulation of the unit cube, with disjoint relative interiors.

Lemma 7.1. *For every finite poset P on X,*

$$\mathcal{O}(P) = \bigcup_{\sigma \in L(P)} \Delta_\sigma.$$

Proof. If σ is a linear extension of P, then $i <_P j$ implies that i occurs before j in σ, so every point of Δ_σ satisfies $x_i \leq x_j$. Thus $\Delta_\sigma \subseteq \mathcal{O}(P)$.

Conversely, every point $x \in [0,1]^X$ lies in at least one simplex of the standard triangulation, obtained by ordering the coordinates and breaking ties arbitrarily. If $x \in \mathcal{O}(P)$, the tie-breaking can be chosen so that $i <_P j$ places i no later than j. The resulting word is a linear extension of P, and x lies in the corresponding Δ_σ. □

For $a \in S_X$, let T_a be the coordinate permutation defined by

$$(T_a x)_{a(i)} = x_i \qquad (i \in X).$$

Then $T_a(\Delta_\sigma) = \Delta_{a\sigma}$, where a acts on the labels in the word. We write

$$a\mathcal{O}(P) := T_a(\mathcal{O}(P)) = \bigcup_{\sigma \in L(P)} \Delta_{a\sigma},$$

for this coordinate-relabeling of $\mathcal{O}(P)$ by a.

Proposition 7.2 (Extension tilers give cube tilings)**.** *Suppose that P is an extension tiler on X, so that*

$$S_X = \bigsqcup_{a \in A} aL(P).$$

Then the coordinate-permuted order polytopes $a\mathcal{O}(P)$ tile the unit cube:

$$[0,1]^X = \bigcup_{a \in A} a\mathcal{O}(P),$$

with overlaps only along faces of the standard simplex triangulation.

Proof. By Lemma 7.1, the pieces $a\mathcal{O}(P)$ are unions of the standard cube simplices indexed by the words in $aL(P)$. The assumed disjoint union $S_X = \bigsqcup_{a \in A} aL(P)$ assigns each standard simplex to exactly one piece. Since the standard simplices tile the cube and have disjoint relative interiors, the stated cube tiling follows. □

Corollary 7.3 (The six-element counterexample order-polytope cube tiling)**.** *For the poset P of Theorem 6.1, called $P038$ in the supplementary enumeration, the unit cube $[0,1]^6$ is tiled, up to shared triangulation faces, by the 48 coordinate permutations $a\mathcal{O}(P)$ with $a \in A = HK \sqcup H\tau K$. Each copy has volume $15/720 = 1/48$.*

Proof. Theorem 6.1 gives $S_6 = \bigsqcup_{a \in A} aL(P)$ with $|A| = 48$ and $|L(P)| = 15$. Apply Proposition 7.2. The volume statement follows either from the 15 standard simplices in $\mathcal{O}(P)$ or from the 48-piece cube tiling. □

7.1 Certified geometry of the P038 tile

The tiling proof above uses only the chamber partition. We also record a small amount of certified geometry for the single tile $\mathcal{O}(P038)$, so that the shape used in the corollary is independently inspectable. The standard order-polytope coordinates in this section satisfy $x_i \le x_j$ when $i <_P j$. For the finite face data it is convenient to use order-ideal indicator coordinates $y_i = 1 - x_i$, in which a cover $i <_P j$ gives $y_j \le y_i$. This integral affine change of coordinates does not change the face lattice, lattice-normalized volume, or Ehrhart data.

For the cover relations

$$0 < 1, \qquad 1 < 2, \qquad 1 < 3, \qquad 4 < 3, \qquad 4 < 5, \qquad 5 < 2,$$

the irredundant natural support inequalities in the y-coordinates are

$$\begin{array}{lll} 0 \le y_2, & 0 \le y_3, & y_0 \le 1, \\ y_4 \le 1, & y_1 \le y_0, & y_2 \le y_1, \\ y_3 \le y_1, & y_3 \le y_4, & y_5 \le y_4, \\ y_2 \le y_5. & & \end{array}$$

Proposition 7.4 (Certified $P038$ order-polytope data)**.** *A finite replay verifies that $\mathcal{O}(P038)$ has 13 vertices, 10 irredundant natural support facets, and boundary f-vector*

$$(f_0, f_1, f_2, f_3, f_4, f_5) = (13, 50, 88, 81, 40, 10).$$

The top face count is $f_6 = 1$. Its Ehrhart h^*-polynomial is

$$h^*_{\mathcal{O}(P038)}(t) = 1 + 6t + 7t^2 + t^3,$$

equivalently

$$L_{\mathcal{O}(P038)}(m) = \binom{m+6}{6} + 6\binom{m+5}{6} + 7\binom{m+4}{6} + \binom{m+3}{6}.$$

Proof. The script `scripts/verify_p038_order_polytope_geometry.py` enumerates the 13 order ideals, forms their 0/1 vertices, checks the ten support facets above, enumerates all nonempty intersections of those facet hyperplanes, and computes the affine dimension of each resulting face. It also counts order-preserving maps $P038 \to \{0, \ldots, m\}$ for $m = 0, \ldots, 6$, derives the h^*-coefficients, and checks the displayed binomial-basis Ehrhart formula. These computations are supporting tile data; the cube tiling itself is still proved by the chamber partition in Proposition 7.2. □

Remark 7.5. This is a coordinate-permutation tiling inherited from the braid-arrangement triangulation. It is not a statement about arbitrary Euclidean congruent copies, and it is independent of any three-dimensional projected model used for visualization.

8 Audit-Level S_6 Laboratory Status

The counterexample in Theorem 6.1 changes the six-element problem from a converse problem into a classification problem. The accompanying laboratory therefore separates three levels:

1. paper-grade positive examples, such as the double-coset construction above;
2. compact obstruction families under development;
3. finite audit data whose current certificates are still too bulky for the main text.

The current audit of the non-series-parallel six-element types satisfying the necessary divisibility condition gives the following status. Here "unresolved" means unresolved inside the current finite laboratory pipeline; it does not mean that every negative row has a short independently checkable paper certificate.

non-series-parallel divisible candidates	57,
non-series-parallel tilers	19,
obstruction-side candidates classified negative by the audit	38,
audit-unclassified survivors	0,
negative audit rows still lacking compact independent certificates	5.

This table is included as audit-level laboratory status data, not as a theorem of this paper. We do not promote the 19/38 split to a theorem because five obstruction-side rows do not yet have short independently checkable certificates in the text or supplement.

Positive structure

The positive rows in the current six-element audit include multiplier sets with subgroup, coset, and double-coset structure. Theorem 6.1 is the cleanest instance: the multiplier set is

$$A = HK \sqcup H\tau K,$$

with $H \cong D_4$ and $K \cong S_3$. The displayed generators suggest a right-hand matching pattern in the cover graph of the poset, while the left-hand symmetry organizes the target labels. This suggests, but does not prove, that the positive side has a group-theoretic organization beyond raw exact-cover search.

Obstruction grammar

On the negative side, many obstruction-side candidates fall into reusable families. The current grammar includes:

terminal-block divisibility, endpoint-pair obstructions,
localized $(3, 3, 2)$ endpoint obstructions, positional gcd obstructions,
single-position balance, ordered-pair balance,
mod-3 peak/threshold obstructions in low partition modules.

These families are valuable because they explain failures by small statistics of words, terminal blocks, or low-dimensional permutation modules, rather than by a raw exact-cover search. The language of low partition modules is the same representation-theoretic environment behind partition-transitivity tests for sets of permutations [MS06, FX26].

The five hard obstruction cases

Five obstruction-side candidates remain the main compression problem:

001, 013, 017, 019, 020.

These are the canonical IDs used in `certificates/s6_classification_status.json`. The canonical ordering rule is part of the laboratory enumeration data rather than the present paper text, so the IDs should be read as stable project-local handles.

The current status table records them as defect-one residual packing obstructions. The numerical rows below are taken from `s6_classification_status.json`. After forcing the identity translate, the audit computation reports that the residual packing graph has maximum clique one below the size required for a tiling. In the table below, "req." is the number of tiles required for a full tiling, "target" is the residual clique size after the identity translate is fixed, and "conf." is the number of binary incompatibility constraints in the residual graph:

cand.	$\lvert L(P)\rvert$	req.	target	max	verts.	conf.
001	48	15	14	13	488	42850
013	24	30	29	28	313	6554
017	24	30	29	28	314	6498
019	24	30	29	28	314	6498
020	24	30	29	28	313	6554

Equivalently, in the audit data, each maximum near-packing leaves a residual set of exactly one tile-size, but the residual cannot be completed by another translate. The laboratory also sees these cases as integer holes in ordered-triple marginals: the linear relaxation is feasible, but the integer decomposition is infeasible.

The current paper does not promote these five rows to standalone theorems. A satisfactory upgrade would be one of the following:

- a compact coloring or clique-cover certificate for the residual packing graph;
- an orbit-quotient certificate under the stabilizer of the identity translate;
- a residual-profile lemma showing that every optimal near-packing leaves a set of tile cardinality that is never a translate of $L(P)$;
- a proof-logged pseudo-Boolean or SAT certificate with an independent verifier.

Until such certificates are extracted, the safe statement is that the current finite audit records a 19/38 laboratory status split among the 57 divisible non-series-parallel candidates, with five negative rows still awaiting compact independent certificates. The theorem-level conclusion of this paper is the explicit failure of the series-parallel converse.

9 Open Problems and Certificate-Compression Targets

Theorem 6.1 shows that the naive converse to the series-parallel tiling theorem is false. The correct classification problem is therefore broader.

Problem 9.1. *Classify the posets P for which $L(P)$ tiles S_n by left translates.*

The positive theorem for series-parallel posets remains valid, but six-element examples show that the class of extension tilers is strictly larger. A useful first target is to describe the additional tilers in S_6 by intrinsic structure rather than by exact-cover witnesses.

Problem 9.2. *Give a paper-grade conceptual classification of the six-element non-series-parallel extension tilers, upgrading the current finite audit into short structural certificates.*

The laboratory status in the previous section suggests that the six-element case is finite and structured: many non-series-parallel types are excluded by local fiber-counting, positional, packing, or low-dimensional permutation-module obstructions, while positive examples often have subgroup, coset, or double-coset complements. The double-coset witness in Theorem 6.1 is the main compact example of this positive structure used in this paper.

Problem 9.3. *Find a conceptual obstruction theory for non-tilers.*

The S_5 computation already shows that cardinality is not enough: $|L(P)|$ may divide $n!$ even when P is not an extension tiler. In S_6, several non-tilers appear to be best understood through packing graphs of left translates, or equivalently through difference-set conditions

$$A^{-1}A \subseteq D \cup \{e\}, \qquad D = \{g \in S_n : L(P) \cap gL(P) = \emptyset\}.$$

The immediate remaining challenge is to turn the five defect-one six-element packing obstructions listed above into short mathematical lemmas or independently verifiable proof certificates.

Problem 9.4. *Develop the analogous extension-tiler theory for other finite Coxeter groups.*

The type-A case is special because chambers are total orders and reflecting halfspaces are precedence constraints. Other Coxeter types may require a different replacement for posets and linear extensions. The separable-element/splitting viewpoint in Weyl groups [GG20] is one natural source of comparison, although the arbitrary left-translate complements allowed here are more flexible than canonical weak-order splittings.

A Profile Quotient Check

This appendix supplies the quotient check used in Lemma 5.2. For a five-element poset Q, define the isomorphism invariant

$$\sigma(Q) = \{(|\{z : z < q\}|,\ |\{z : q < z\}|) : q \in Q\},$$

as a multiset. The table lists the 15 quotient classes of one-point N-extension profiles, together with $|L|$ and σ. Thus, for example, $(0,4)$ denotes one element with no lower elements and four upper elements. The pair $(|L|, \sigma)$ is different in every row, so no further profile identifications occur beyond the five listed in Lemma 5.2.

(D, U)	$\lvert L\rvert$	σ
$(\emptyset, abcd)$	5	$(0,4), (1,1), (1,2), (2,0), (3,0)$
$(abcd, \emptyset)$	5	$(0,2), (0,3), (1,1), (2,1), (4,0)$
$(\emptyset, bcd) \cong (c, bd)$	7	$(0,1), (0,3), (1,2), (2,0), (3,0)$
$(ac, b) \cong (abc, \emptyset)$	7	$(0,2), (0,3), (1,0), (2,1), (3,0)$
$(\emptyset, abd) \cong (acd, \emptyset)$	8	$(0,2), (0,3), (1,1), (2,0), (3,0)$
$(\emptyset, ab) \cong (a, b)$	9	$(0,2), (0,2), (1,0), (1,1), (3,0)$
$(c, d) \cong (cd, \emptyset)$	9	$(0,1), (0,3), (1,1), (2,0), (2,0)$
(c, b)	11	$(0,1), (0,3), (1,0), (1,1), (3,0)$
$(\emptyset, bd)$	14	$(0,1), (0,2), (0,2), (2,0), (3,0)$
$(ac, \emptyset)$	14	$(0,2), (0,3), (1,0), (2,0), (2,0)$
$(\emptyset, d)$	16	$(0,1), (0,1), (0,2), (2,0), (2,0)$
$(a, \emptyset)$	16	$(0,2), (0,2), (1,0), (1,0), (2,0)$
$(\emptyset, b)$	18	$(0,1), (0,1), (0,2), (1,0), (3,0)$
$(c, \emptyset)$	18	$(0,1), (0,3), (1,0), (1,0), (2,0)$
$(\emptyset, \emptyset)$	25	$(0,0), (0,1), (0,2), (1,0), (2,0).$

B Computational Audit

This appendix records the independent computational audit for Theorem 5.6. The theorem itself is proved in the main text by finite classification and fiber counts. The current repository snapshot stores the expected audit output in

`certificates/s5_expected_output.txt`.

The current replay script

`scripts/replay_s5_extension_tiler_audit.py`

reproduces the aggregate S5 counts and the former rank and packing audit certificates. These computations are retained as independent audit support; the proof of Theorem 5.6 remains the finite table plus the elementary obstructions in the main text.

The six-element counterexample in Theorem 6.1 has a separate self-contained verifier:

`scripts/verify_s6_p038_biset_counterexample.py`.

It reconstructs the poset, the subgroups H, K, the multiplier set $A = HK \sqcup H\tau K$, and checks that the 48 translates of the 15-element tile cover all 720 elements of S_6 exactly once.

The broader six-element laboratory status is recorded in

`certificates/s6_classification_status.md`

and in the companion JSON file with the same base name. The five hard residual-clique rows are retained only as bounded computational status and are not used as proof of a theorem in this paper.

Aggregate counts

The script verifies:

labeled posets	4231,
unlabeled posets	63,
series-parallel labeled posets	2791,
series-parallel unlabeled posets	48,
divisible labeled posets	3151,
divisible unlabeled posets	51,
non-series-parallel divisible labeled posets	360,
non-series-parallel divisible unlabeled posets	3.

The former rank obstructions

For the first $|L(P)| = 5$ type, one representative has transitive relations

$$(0,1),(0,2),(0,3),(0,4),\\ (1,2),(1,3),(4,2).$$

Its linear extensions are

$$(0,1,3,4,2),\quad (0,1,4,2,3),\quad (0,1,4,3,2),\\ (0,4,1,2,3),\quad (0,4,1,3,2).$$

The certificate is

$$\mathrm{rank}_{\mathbb{F}_5}(M) = 115, \qquad \mathrm{rank}_{\mathbb{F}_5}(M \mid \mathbf{1}) = 116.$$

For the second $|L(P)| = 5$ type, one representative has transitive relations

$$(0,1),(0,2),(0,3),(1,2),(3,2),(4,1),(4,2).$$

Its linear extensions are

$$(0,3,4,1,2),\quad (0,4,1,3,2),\quad (0,4,3,1,2),\\ (4,0,1,3,2),\quad (4,0,3,1,2).$$

The same rank certificate holds:

$$\mathrm{rank}_{\mathbb{F}_5}(M) = 115, \qquad \mathrm{rank}_{\mathbb{F}_5}(M \mid \mathbf{1}) = 116.$$

The former packing obstruction

For the $|L(P)| = 8$ type, one representative has transitive relations

$$(0,1),(0,2),(0,3),(1,2),(4,2),(4,3).$$

Its linear extensions are

$$(0,1,4,2,3),\quad (0,1,4,3,2),\quad (0,4,1,2,3),\quad (0,4,1,3,2),\\ (0,4,3,1,2),\quad (4,0,1,2,3),\quad (4,0,1,3,2),\quad (4,0,3,1,2).$$

The translate family has 120 distinct left translates. Among the translates disjoint from the base translate there are 96 candidates, and the maximum packing containing the base translate has size 14. Since a tiling would require 15 translates, this type is not an extension tiler.

Data and Code Availability

The public repository is:

`https://github.com/aconsciousfractal/type-a-poset-cones-and-extension-tilers`

The reproducibility boundary for this snapshot is recorded in `REPRODUCE.md`, with per-file hashes in `MANIFEST_SHA256.txt`. The immutable public snapshot is release `v0.2.1`. The theorem-level six-element counterexample is checked by `scripts/verify_s6_p038_biset_counterexample.py`. The order-polytope cube-tiling corollary is a deterministic consequence of the same exact factorization and does not depend on any projected visual artifact. The certified finite replay data for Proposition 7.4 are checked by `scripts/verify_p038_order_polytope_geometry.py` and recorded in `certificates/p038_order_polytope_geometry.json`. The five-element finite audit can be replayed with `scripts/replay_s5_extension_tiler_audit.py`. The broader six-element laboratory table is stored as audit-level data in `certificates/s6_classification_status.md` and `certificates/s6_classification_status.json`. The five hard rows are bounded computational status and are not used as proof of a theorem in this paper.

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